# Basic HTML Tags

**1. Introduction to HTML**

- HTML stands for **HyperText Markup Language**. It is the standard language used to create and structure web pages. HTML uses **tags** to define elements such as headings, paragraphs, images, links, tables, and forms.
- A basic HTML document tells the browser:
  - What content to display
  - How the content is structured

**2. Basic Structure of an HTML Document**

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

**Explanation:**

- <!DOCTYPE html> → Declares HTML5 document
- <html> → Root element of HTML
- <head> → Metadata (title, styles, scripts)
- <title> → Title shown on browser tab
- <body> → Visible content of the page

**3. HTML Tags and Elements**

An **HTML tag** is enclosed in angle brackets < >.

Example:

```
<p>This is a paragraph</p>
```

-  → Opening tag
-  → Closing tag

**4. Heading Tags**

HTML provides **six levels of headings**, from <h1> (largest) to <h6> (smallest).

```
<h1>Main Heading</h1>
<h2>Sub Heading</h2>
<h3>Section Heading</h3>
<h4>Subsection</h4>
<h5>Minor Heading</h5>
<h6>Smallest Heading</h6>
```

- **Use headings to structure content**, not just for styling.

**5. Paragraph Tag **

Used to define paragraphs of text.

```
<p>HTML is easy to learn.</p>
<p>It is used to create web pages.</p>
```

**6. Line Break and Horizontal Rule**

**Line Break <br>**

Breaks a line without starting a new paragraph.

```
HTML<br>CSS<br>JavaScript
```

**Horizontal Rule <hr>**
Draws a horizontal line.
<hr>
**7. Formatting Tags**

| Tag | Description | Example |
|---|---|---|
|  | Bold text | Bold |
|  | Important text | Important |
|  | Italic text | Italic |
|  | Emphasized text | Emphasis |
| <u> | Underline | <u>Underline</u> |
| <mark> | Highlight | <mark>Marked</mark> |

Example:
HTML is very important for <mark>Web Development</mark>.
**8. Anchor Tag <a> (Links)**
Used to create hyperlinks.
<a href="https://www.google.com">Visit Google</a>
**Open link in new tab:**
<a href="https://www.google.com" target="_blank">Google</a>
**9. Image Tag <img>**
Used to display images.
<img src="image.jpg" alt="Sample Image" width="200" height="150">

- <img> is a **self-closing tag**.

**10. List Tags**
**Ordered List <ol>**

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

**Unordered List <ul>**

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
  <li>Mango</li>
</ul>
```

**11. Table Tags**
Used to display data in tabular form.

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Ravi</td>
    <td>20</td>
  </tr>
</table>
```

**Table Tags:**

- <table> → Table
- <tr> → Table row
- <th> → Table heading
- <td> → Table data

**12. HTML Comments**

Comments are not displayed in the browser.

<!-- This is a comment -->

**Practice Questions**

1. Write an HTML program to display your **name, roll number, and department**, each on a new line using appropriate tags.
2. Write an HTML program to display a heading "Web Technologies" followed by a paragraph describing HTML.
3. Write an HTML program that uses <h1> to <h6> tags to display their names.
4. Write an HTML program to insert a horizontal rule after every paragraph.
5. Write an HTML program to display a sentence where one word is **bold**, one is *italic*, and one is underlined.
6. Write an HTML program to create an unordered list of **programming languages** you know.
7. Write an HTML program to create an ordered list of **steps to prepare for an exam**.
8. Write an HTML program to create a webpage with two links:
    - One link opens Google in the same tab
    - Another link opens YouTube in a new tab
9. Write an HTML program to display a nested list (list inside another list).